DE–6211 **11**

DISTANCE EDUCATION

M.Sc. (IT) DEGREE EXAMINATION, MAY 2013.

PRINCIPLES OF INFORMATION TECHNOLOGY

(2002 onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)

Answer ALL the questions.

1. What is word processing?

2. What are specialised software?

3. What is Intranet?

4. What are the factors affecting data transmission?

5. What is meant by structured programming?

6. What are I/O streams in C?

7. Differentiate arrays and pointers.

8. How do you store multidimensional arrays in Memory?

9. What is inline function?

10. What is recursion?

PART B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Explain the features of various Internet web browsers.

12. Explain briefly about

    (a) Database Software

    (b) Presentation graphics software.

13. Explain briefly about Electronic Data Interchange.

14. Explain with examples, various operators used in C language.

15. Write a program in C to define a structure and process the same with an example.

16. Explain the concept of Macros.

PART C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Describe the evolution of ICT.

18. Discuss in detail about cyber ethics.

19. Explain with examples, the concepts of Pass-by-value and Pass-by-reference in C.

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DE–6212 **12**

DISTANCE EDUCATION

M.Sc. (IT) DEGREE EXAMINATION, MAY 2013.

OPERATING SYSTEMS

(2002 onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)

Answer ALL questions.

1. List out the four components in operating system.
2. What is a spooling?
3. What are the different states in process?
4. Define semaphores.
5. Give the purpose of device drives.
6. What is meant by directed graph?
7. What is meant by dynamic linking?
8. Define paging.
9. What are the different access methods in files?
10. Write any three file access types.

PART B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Briefly explain about multiprogrammed batch system.
12. Explain the first come, first served (FCFS) scheduling with an example.

13. Explain Banker's algorithm.

14. Briefly discuss about the swapping.

15. Explain the LRU algorithm with example.

16. Briefly discuss about contiguous allocation of disk space.

PART C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Explain in detail process state life cycle.

18. Discuss about the deadlock avoidance.

19. Explain directory structure in detail.

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DE–6213 **13**

DISTANCE EDUCATION

M.Sc.(IT) DEGREE EXAMINATION, MAY 2013

OBJECT ORIENTED PROGRAMMING AND C++

(2002 Onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)

Answer ALL questions.

1. What is meant by data hiding?

2. What is a class?

3. List the various sub classes of stream class.

4. Briefly explain the random access file operations in C++.

5. Define the term Constructor.

6. What is meant by function overloading?

7. Define pointers.

8. Explain implicit and explicit type conversion.

9. What is polymorphism?

10. Explain the concept of multilevel inheritance.

PART B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. List the salient features of object oriented programming.

12. Explain the various data types available in C++.

13. Discuss on any two types of looping statements available in C++.

14. Explain the recursive function with suitable example.

15. Write a C++ program to read and write a text content from/ to file.

16. How to reuse the code with member objects?

PART C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Explain the various types of operators available in C++. Give its hierarchy.

18. Write a program in C++ to overload '+' (Plus) operator.

19. Define the term inheritance. Discuss the various types of inheritance.

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DE–6214 **14**

DISTANCE EDUCATION

M.Sc. (I.T.) DEGREE EXAMINATION, MAY 2013.

DATA STRUCTURES AND ALGORITHMS

(2002 onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)

Answer ALL questions.

1. What are primitive and composite data types?
2. What is data structure?
3. Write the applications of stack and queue.
4. List the operations that can be performed on strings.
5. Define Binary Tree.
6. List the applications of Trees.
7. What is heap?
8. Distinguish between linear and binary search.
9. Convert (A+B) × C/D into postfix notation.
10. How will you convert a forest into Binary Tree?

PART B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. What is an algorithm? What are the desirable characteristics of an algorithm? What are space and time complexity of algorithms?

12. Write about composite functions in string manipulation and pattern matching.

13. Write the algorithm for evaluation of an expression using stack and explain it with an example.

14. Explain the Graph Traversal algorithms.

15. Write the algorithm for selection sort and explain it with an example.

16. Explain the concept of 'hashing'.

PART C — ($2 \times 15 = 30$ marks)

Answer any TWO questions.

17. (a) What is stack? Explain the various operations on stack.

    (b) What is Queue? Explain the various operations on Queue.

18. Explain the different binary tree traversal methods.

19. Write the algorithm for heap sort and explain it.

DE–6215 **15**

DISTANCE EDUCATION

M.Sc. (IT) DEGREE EXAMINATION, MAY 2013.

OBJECT ORIENTED DBMS

(2002 onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)

Answer ALL questions.

1. Define Table and its entries.
2. Define Transaction.
3. Explain class hierarchy.
4. What do you mean by database queries and updates?
5. Explain about Triggers.
6. Define Union, Intersection with example.
7. Define loss-less join with example.
8. Define physical database design.
9. Define Access control.
10. Define Object identity.

PART B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Explain the structure of database system.
12. Discuss Key Constraints in detail.
13. Write short notes on Aggregate operators.

14. Explain indexing technique in detail.

15. Explain storing data in a distributed DBMS in detail.

16. Write short notes on Querying SML data.

PART C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. (a) How to describe and store data in database? Explain.

    (b) Write notes on Dynamic SQL.

18. (a) Define Normalization. How to design a database using normalization technique.

    (b) Explain XML architecture in detail.

19. Explain Distributed Query Processing in detail.

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DE–6216 **21**

DISTANCE EDUCATION

M.Sc. (IT) DEGREE EXAMINATION, MAY 2013

INTERNET PROGRAMMING AND WEB DESIGN

(2002 onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)

Answer ALL questions.

1. What are the advantages of Internet?

2. What is network?

3. Write short notes on Access Methods.

4. What is a Datagram?

5. What is the purpose of Cookies?

6. What is an URL?

7. What is CGI?

8. Write the steps involved in displaying a HTML document.

9. Write a note on modifiers in Perl.

10. How can you create frame window in Java applet?

PART B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. What are the different types of Network?

12. Explain the following:

    (a) Browser (5)

    (b) Down loading, uploading (5)

13. Explain briefly about TCP/IP protocol model with neat diagram.

14. Discuss in detail about subroutines in Perl language.

15. Write short notes on following:

    (a) <HR> (5)

    (b) <PRE> (5)

16. Write short notes on :

    (a) SGML (5)

    (b) Shockwave. (5)

PART C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. (a) Explain the three types of links that you can create in HTML with example.

    (b) Explain about the various types of LAN topology.

18. (a) Explain about VDU live technology.

    (b) Explain about Hyperlinks.

19. (a) Comment on VB script control structure.

    (b) Write a note on how function accept the arguments. Explain with an example.

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DE–6217 **22**

DISTANCE EDUCATION

M.Sc. (IT) DEGREE EXAMINATION, MAY 2013.

COMPUTER NETWORKS

(2002 Onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)

Answer ALL questions.

1. What is meant by LAN? How does it differ from other networks?

2. What is a protocol? Give examples.

3. What are the three major problems associated with transmission lines?

4. What is piggy backing?

5. What is a Petrinet? What are its elements?

6. How does SVC (Switched Virtual Circuit) differ from PVC (Permanent Virtual Circuit)?

7. List out any two interfaces used in ATM networks.

8. When is upward multiplexing used?

9. What is meant by authentication?

10. List out any five generic domain names used in internet.

PART B — ($4 \times 10 = 40$ marks)

Answer any FOUR questions.

11. Write a note on network topologies.

12. Explain the various categories of guided media.

13. Narrate on the key points to be remembered in allocating channels.

14. Explain the functionality of PPP in data transmission.

15. Outline the various elements of transport protocols.

16. How does digital signatures help us in secured data transmission?

PART C — ($2 \times 15 = 30$ marks)

Answer any TWO questions.

17. Explain the TCP/IP network reference model in detail.

18. Discuss the methodology behind the ALOHA protocols.

19. Write an essay on Email and WWW.

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DE–6218 **23**

DISTANCE EDUCATION

M.Sc. (IT) DEGREE EXAMINATION, MAY 2013.

SOFTWARE ENGINEERING

(2002 onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)

Answer ALL questions.

1. Write about the different software engineering layers.

2. Which is more important in software engineering the product or the process?

3. What is meant by problem decomposition?

4. Define scheduling. Give two examples.

5. What is risk management? Write any two activities.

6. What is SQA? Write any two applications.

7. What are the basic elements of a system?

8. Explain the term 'coupling'.

9. What is system testing?

10. What is the purpose of economic analysis in software development?

PART B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Explain the prototype model in software engineering.

12. Briefly explain about Make/Buy decision in software design.

13. Write short notes on :

    (a) Economic analysis

    (b) Technical analysis.

14. What are the elements of analysis model? Explain how to create analysis model.

15. What are the various types of architectural styles? Explain.

16. Explain the testing strategy of software engineering.

PART C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Explain how can you estimate the scope of software and resources.

18. What is meant by software design? What are the design principles are followed in software design?

19. Explain :

    (a) White box-testing

    (b) Control structure testing

    (c) Integration testing.

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DE–6219 **24**

DISTANCE EDUCATION

M.Sc. (IT) DEGREE EXAMINATION, MAY 2013.

VISUAL PROGRAMMING

(2002 onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)

Answer ALL questions.

1. What are the data types in windows programming?

2. What is context help? Give an example.

3. What do you mean by MDI forms?

4. With its general format, explain the statement in VB for decision making.

5. Give an example for the method of using option buttons.

6. Mention any six important properties of text boxes.

7. List out the VC++ components.

8. What do you understand by event handling?

9. What is ODBC?

10. What is the use of DLLs?

PART B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Explain the different traditional programming paradigms.

12. Explain the looping statements in VB with suitable examples.

13. Explain how menus can be incorporated in VB applications. Give an example.

14. Explain the following in VB : Grid controls, dialog boxes.

15. Explain how keyboard events are handled in VC++ with an example.

16. Explain the facilities available in VC++ for debugging.

PART C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Write notes on the following :

    (a) Windows programming

    (b) SDK tools.

18. Discuss MFC file handling in VC++.

19. Explain the following :

    (a) Recordset objects in VB

    (b) OLE in VC++.

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DE–6220 **25**

DISTANCE EDUCATION

M.Sc. (IT) DEGREE EXAMINATION, MAY 2013.

MULTIMEDIA AND APPLICATIONS

(2002 onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)

Answer ALL questions.

1. What is multimedia? Why do we need it?
2. Expand the followings : (a) ISDN and (b) PCI.
3. What do you understand by multispectral images?
4. What is tweening?
5. Why do we need compression?
6. Define Vectorisation.
7. What is resolution? What are its types?
8. What is Blob?
9. What do you understand by DVI?
10. Name the standards followed in transmitting the multimedia documents on the net.

PART B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Explain the different types of development tools.
12. Discuss the various standards for developing multimedia documents.

13. Narrate on colour models.

14. How does the framework provide a unified picture on multimedia communication and information processing?

15. Present a note on the multimedia interaction with single user.

16. Outline the use of OO languages in developing MM presentation.

PART C — ($2 \times 15 = 30$ marks)

Answer any TWO questions.

17. Write an essay on the application of multimedia in training and education.

18. Describe the various video compression techniques.

19. Explain the role of CBRS in multimedia presentations.

DE–6623 **16**

DISTANCE EDUCATION

M.Sc. (IT) DEGREE EXAMINATION, MAY 2013.

C++ LABORATORY

Time : Three hours Maximum : 100 marks

Examiner has to choose any ONE question.

Break-up marks

Record : 10
Algorithm : 10
Program : 50
Debugging : 10
Execution : 10
Result : 10
Total : 100

1. Write a program in C++ to find whether a given string is a palindrome and find the length of the string.

---------- Cut here ----------

2. Write a program in C++ to convert temperature given in Fahrenheit to Celsius and vice-versa.

---------- Cut here ----------

3. Write a program to find the difference and total length of 2 tubes specified in meters and centimeters. Use operator overloading.

4. Write a program in C++ to concatenate two given strings using dynamic constructors.

---------- Cut here ----------

5. Create a class EMPLOYEE that contains Employee number, Employee name and address. Write a menu driven C++ program to get the 'n' number of employee details and display all details in employee namewise sorted order.

---------- Cut here ----------

6. Write a C++ program to create a class, get n names and display them in sorted order. Use pointers.

---------- Cut here ----------

7. Write a menu driven C++ program to add and subtract two matrices of order $m \times n$. Use operator overloading.

---------- Cut here ----------

8. Write a C++ program using polymorphism to calculate the square of any two numbers of type int, float, double and long.

DE–6624 **17**

DISTANCE EDUCATION

M.Sc (IT) DEGREE EXAMINATION, MAY 2013.

DATA STRUCTURES LABORATORY

(2002 onwards)

Time : Three hours Maximum : 100 marks

Break-up of marks :

Record : 10
Algorithm : 10
Program : 50
Debugging : 10
Execution : 10
Result : 10
Total : 100

Instruction :

Examiner has to choose TWO questions One from Section – A and another from Section – B for each candidate.

Each question carries 50 marks

Candidate has to write algorithm and C++ code.

SECTION A

1. Write a C++ program to find the inverse and transpose of the matrix.

---------- Cut here ----------

2. Write a C++ program for searching element using binary search method.

3. Write a C++ program to sort the following list using selection sort.

   34 12 54 9 8 67 2 45

---------- Cut here ----------

4. Write a C++ program for creation of linked list and performing its operations using array.

---------- Cut here ----------

5. Write a C++ program to sort elements using quick sort.

---------- Cut here ----------

SECTION B

6. Write a C++ program for searching elements using linear search method.

---------- Cut here ----------

7. Write a C++ program for linked list implementation of queue.

---------- Cut here ----------

8. Write a C++ program to copy the content of one file into another file.

9. Write a C++ program for creation of linked list and performing its operations using pointers.

---------- Cut here ----------

10. Write a C++ program to sort the elements using bubble sort.

DE–6625 **26**

DISTANCE EDUCATION

M.Sc. (IT) DEGREE EXAMINATION, MAY 2013.

INTERNET PROGRAMMING LABORATORY

Time : Three hours Maximum : 100 marks

Examiner has to choose any ONE question.

Break-up marks

Record Note Book	: 10
Algorithm	: 10
Program	: 50
Debugging	: 10
Execution	: 10
Result	: 10
Total	: 100

1. (a) Write a Java Script to create an order form to select the home appliances (home need products).

   (b) Design a web page using HTML code to show your biodata.

---------- Cut here ----------

2. (a) Write a VB script to do the following :

   (i) Check the given password.

   (ii) Change the existing password.

   (b) Write a Java program using Applet to create the frames and its controls.

3. (a) Write a Java Script to create a window by using the confirm message.

   (b) Design a web page using HTML code to display the details of your institution where you are doing course.

---------- Cut here ----------

4. (a) Write a VB Script to create a calender for a given month and year.

   (b) Write a Java program using Applet to display any 3 images when 3 buttons in the Border Layout are clicked. The image should be displayed at the center.

---------- Cut here ----------

5. (a) Write a Java Script to create a color palette and display the background in the color chosen from the palette.

   (b) Write a Swing program to create buttons with

      (i) Tool tip text

      (ii) Image

      (iii) Border

      (iv) Short cut key.

---------- Cut here ----------

6. (a) Write a VB Script to do the following :

      (i) Display the current date.

      (ii) Find the difference between the two dates.

   (b) Write a Java Program to use database connectively for displaying the contents of an employee file.

7. (a) Write a VB Script to create a calender for a given month and year.

   (b) Write a Swing Program to create the tabbed panels.

---------- Cut here ----------

8. (a) Write a JavaScript to create an order form to select the home appliances.

   (b) Write an applet program to draw a filled rectangle and circle. Use graphic controls.

---------- Cut here ----------

9. (a) Write a Java Program to create an applet to display the message "HAVE A NICE DAY" with different colors and fonts for each click.

   (b) Design a web page using HTML code to display the details of your institution where you are doing course.

---------- Cut here ----------

10. (a) Write a Java Script to create a color palette and display the background in the color chosen from the palette.

    (b) Design a web page using HTML code shows your biodata.

DE–6626 **27**

DISTANCE EDUCATION

M.Sc. (IT) DEGREE EXAMINATION, MAY 2013.

VISUAL PROGRAMMING – LABORATORY

(2002 Onwards)

Time : Three hours Maximum : 100 marks

Examiner to select and give ONE question to each candidate.

1. (a) Write event procedure in VB to

   (i) Display date and time at run time

   (ii) To find the power of a number.

   (b) Write a Visual C++ program to create a window of desired size using MFC.

---------- Cut here ----------

2. (a) Write event procedures in VB to perform the following

   (i) Reverse and sharing

   (ii) To check whether the given string is palindrome or not.

   (b) Write a visual C++ program to handle windows messages.

3. (a) Using MS Flex Grid control, display the multiplication and addition table of 10 rows and columns, in VB.

(b) Write a Visual C++ program to fill back ground of a client area with a bit map.

---------- Cut here ----------

4. (a) Using control array, create a simple calculator to perform addition, subtraction multiplication and division.

(b) Write a Visual C++ program to get the status of the shift and toggle keys using MFC.

---------- Cut here ----------

5. (a) Using built–in Ax control, develop the windows NOTEPAD with file and edit menu operations.

(b) Write a Visual C++ program to generate a status bar and show the status of caps lock, num lock and scroll lock in it.

---------- Cut here ----------

6. (a) Using remote data access object, implement the operations insertion, search and update with employee records.

(b) Write a Visual C++ program to create a list box in a window.

7. (a) Using MSFlex Grid control, display the multiplication and addition table of 15 rows and columns.

(b) Write VC++ program to find out whether a mouse is attached or not; and if attached how many buttons are present or not.

---------- Cut here ----------

8. (a) Write event procedure to perform the following reversing a string to change the case from lower to upper.

(b) Write a VC++ program to create a window of desired size using MFC.

---------- Cut here ----------

9. (a) Using RDA object, implement the following operations with student marks details

(i) Insertion

(ii) Updation

(iii) Searching.

(b) Write a VC++ program to handle windows messages in MFC program.

---------- Cut here ----------

10. (a) Write about procedures to

(i) display date and time in a label.

(ii) to find the power of a nuclear.

(b) Write a VC++ program to fill background of the client area with a bitmap.